



Comparative analysis of assays to measure CAR T-cell-mediated cytotoxicity

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The antitumor efficacy of genetically engineered ‘living drugs’, including chimeric antigen receptor and T-cell receptor T cells, is influenced by their activation, proliferation, inhibition, and exhaustion. A sensitive and reproducible cytotoxicity assay that collectively reflects these functions is an essential requirement for translation of these cellular therapeutic agents. Here, we compare various in vitro cytotoxicity assays (including chromium release, bioluminescence, impedance, and flow cytometry) with respect to their experimental setup, appropriate uses, advantages, and disadvantages, and measures to overcome their limitations. We also highlight the US Food and Drug Administration (FDA) directives for a potency assay for release of clinical cell therapy products. In addition, we discuss advanced assays of repeated antigen exposure and simultaneous testing of combinations of immune effector cells, immunomodulatory antibodies, and targets with variable antigen expression. This review article should help to equip investigators with the necessary knowledge to select appropriate cytotoxicity assays to test the efficacy of immunotherapeutic agents alone or in combination.

Immunotherapy harnesses a patient’s immune system to combat cancer, either by restoring the function of endogenous immunity or by generating new immune responses through the establishment of synthetic immunity. The former has been achieved through the use of immune checkpoint inhibitors—a class of immunomodulatory antibodies that target inhibitory receptors on T cells, such as programmed cell death protein 1 (PD-1) and cytotoxic T-lymphocyte-associated protein 4 (ref. ¹)—and bispecific T-cell engager (BiTE) antibodies, which mediate T-cell responses by binding to a target on the tumor cell and T cell simultaneously². With continuing advances in genetic engineering, biological therapies, such as adoptive T cells and oncolytic viruses, are being translated to clinical use to lyse tumor cells. The resulting release of cancer cell antigens in an altered tumor immune microenvironment attracts cells of both the innate and the adaptive immune system to the tumor³, promoting neoantigen immune responses⁴.

Adoptive cell therapy relies on the isolation and expansion of the patient’s immune cells (such as T cells), which are genetically modified to express a cancer antigen-specific chimeric antigen receptor (CAR) or T-cell receptor (TCR). CAR T cells—a patient’s own T cells transduced to express a CAR—are called ‘living drugs’ because of their ability to proliferate, expand, and persist following antigen stimulation. They have been approved for the treatment of hematological malignancies. CAR T cells, CAR natural killer (NK) cells, and CAR macrophages, all subsets of engineered immune cells, are currently in clinical trials for the treatment of solid tumors⁵. Genetic engineering tools, such as clustered regularly interspaced short

palindromic repeats (CRISPR)⁶, can be used to potentiate CARs by targeting specific pathways involved in the suppression of the immune effector function in the tumor microenvironment⁷. More importantly, such tools can be used to insert CARs at the TCR α constant locus⁸. Following the establishment of the feasibility, safety, efficacy, and persistence of CRISPR-engineered T cells in patients⁹, the use of multiplex genome engineering techniques is being extended to allogeneic donor T cells, thereby increasing their broad applicability.

Given the variety and complexity of multigene-engineered living drugs, there is an increasing need for reproducible and high-throughput screening assays to select the best therapeutic agents. A first step to screening agents is to measure the cancer cell-killing ability of an effector cell with a cytotoxicity assay. In this review article, we describe assays that quantify target cell lysis mediated by effector cells as a measure of cell-mediated cytotoxicity. We specifically focus on four of the most commonly used assays to investigate cell-mediated cytotoxicity: the chromium (⁵¹Cr)-release assay (⁵¹Cr assay), the luciferase-mediated bioluminescence imaging (BLI) assay, the impedance-based assay, and the flow cytometry assay (flow assay) (Fig. 1 and Table 1). These assays differ in their requirements for target cell labeling, culture time, principal measures of cytotoxicity, number of measurements (temporal or endpoint), ability to measure differential cytotoxicity on heterogeneous targets, throughput, and automatability. We further discuss their advantages and limitations and give specific examples of their applications for different classes of immunotherapies (Tables 2–5).

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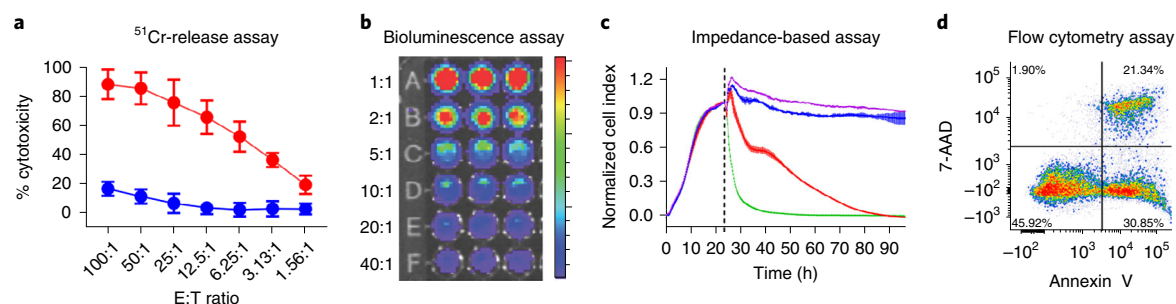


Fig. 1 | Interassay comparison of cell-mediated cytotoxicity assessment. **a–d**, Representative readouts of the respective assays to quantify cell-mediated cytotoxicity. **(a)** ⁵¹Cr-release assay: ⁵¹Cr release of labeled target cells cocultured with target-specific effector cells (red curve) or control cells (blue curve) at different effector-to-target (E:T) ratios determined relative to a maximum and a spontaneous ⁵¹Cr-release control; **(b)** bioluminescence assay: bioluminescence intensity proportional to the number of viable luciferase-expressing target cells, indicative of effector cell-mediated cytotoxicity at different E:T ratios; **(c)** impedance-based assay: detachment of adherent target cells, monitored as a decrease in normalized cell index upon addition of target-specific effector cells (red curve) or control cells (blue curve) at a specific E:T ratio and defined time after seeding of target cells (dotted line) for untreated target cells (purple curves) and target cells treated with 0.1% Triton X-100 (100% lysis control, green curves); **(d)** flow cytometry assay: 7-aminoactinomycin D (7-AAD) and annexin V staining of target cells undergoing effector cell-mediated apoptosis and cell death (dot plot), showing live cells (lower left), early apoptotic cells (lower right), and cells at terminal stage of apoptosis or cell death (upper right).

⁵¹Cr assay

Historically, methods to measure cell-mediated cytotoxicity have been based on the quantification of target cells with loss of membrane integrity. The ⁵¹Cr assay, which works on this principle, was first described more than five decades ago by Brunner et al.¹⁰ and gained popularity as the gold standard to measure T-cell- and NK-cell-mediated cytotoxicity^{11,12}.

In this assay, target cells are labeled with radioactive chromium (⁵¹Cr) and coincubated with effector cells at different effector-to-target (E:T) ratios. During effector-mediated cell killing, the target cells lose their membrane integrity and release ⁵¹Cr into the culture medium. The radioactivity (measured in the supernatant using a gamma scintillator counter) is proportional to the number of target cells killed during that period. A spontaneous release control (⁵¹Cr-labeled target cells alone in culture medium) and a maximum release control (⁵¹Cr-labeled target cells in culture medium treated with a detergent such as Triton X-100) are included to allow determination of the percentage of specific lysis, which is calculated using the following formula: [(experimental release – spontaneous release)/(total release – spontaneous release)] × 100 (refs. ^{10,13}).

The standard ⁵¹Cr assay measures the ⁵¹Cr released 4 h after coincubation of effector and target cells, but longer incubation times might be necessary to allow observation of significant cell killing for more resistant target cells and for solid tumor cells in particular (Table 2). As an endpoint assay, the ⁵¹Cr assay is usually limited to a single time point, but measurements can be obtained at more than one time point if only part of the radioactive supernatant is used¹⁴. However, spontaneous ⁵¹Cr release from the target cells leads to an increased background and a reduced signal-to-background ratio over time, which thereby prevents cytotoxicity measurement over extended periods¹⁵. ⁵¹Cr release in the assay is higher with detergent solubilization than with hypotonic lysis, leading to overestimation of ⁵¹Cr release and underestimation of cytotoxicity¹⁶.

In light of the potential effects of ⁵¹Cr on the health of the investigator, an alternative method that uses nonradioactive chromium was developed in 1995 (ref. ¹⁷). However, because of

its requirement for expensive equipment and specialized staff and its low-throughput capability, this method has never achieved widespread use. Despite its technical limitations, the ⁵¹Cr assay is considered the gold standard for measurement of cell-mediated cytotoxicity. For evaluation of the cytotoxicity of immune cells transduced with newer constructs, such as CARs or TCRs, our laboratory performs the ⁵¹Cr assay as a first step, with inclusion of well-established constructs for comparison. However, not all laboratories possess the mandatory radiation license and infrastructure for radioisotope storage, disposal, and monitoring. The following sections provide investigators with nonradioactive alternatives to the ⁵¹Cr assay.

BLI luciferase assay

This BLI-based cytotoxicity assay measures the light (photons) emitted from target cells that are transduced with a luciferase reporter gene. The luciferase substrate is processed only by live cells that express the luciferase transgene. Consequently, the cytotoxicity of effector cells can be measured by the decrease in the BLI signal¹⁶.

Early luciferase-based cytotoxicity assays measured the activity of secreted firefly luciferase in the supernatant upon lysis of target cells. However, because of the short half-life (~30 min) of firefly luciferase in culture medium, these assays were restricted to assay conditions in which substantial lysis of the target cells occurs within a few hours¹⁸. Furthermore, because of its molecular weight of 62 kDa¹⁹, firefly luciferase might not efficiently leak into the supernatant upon effector cell-mediated pore formation in the target cell²⁰. Therefore, particularly for cytotoxicity studies that require longer incubation times, complete cell lysis might be necessary to allow for measurement of cell-associated luciferase activity²⁰, which would limit the readout to a single endpoint. However, luciferases derived from marine organisms are smaller in size, exhibit longer half-lives, and have increased brightness^{21–23}. For example, nanoluciferase, a 19-kDa reporter protein derived from deep-sea shrimp that has a half-life of 7.7 d in

Table 1 | Properties of cell-mediated cytotoxicity assays

Assay format	Principal measure of cytotoxicity	Radioactive materials needed	Target cell labeling required	Genetic modification of target cells required	Endpoint/kinetic	Real-time measurement	Maximum time point measured	Ability to measure differential cytotoxicity on heterogeneous targets	Throughput and automatability
Chromium release	⁵¹ Cr release	Yes	Yes	No	Endpoint	No	18–24 h	No	Low
Bioluminescence	Luciferase activity	No	No	Yes (reporter gene)	Endpoint	No	Days	No	High
Impedance	Cell detachment	No	No	No	Temporal	Yes	Days	No	High
Flow cytometry	Live/dead staining, phenotype	No	Yes	No	Endpoint	No	Days	Yes	High

extracellular medium, is completely released during cell death, enabling measurement of the luciferase signal in supernatants over prolonged periods²¹. In a similar assay, sensitivity was not compromised when a centrifugation step to separate supernatant and cell pellet was omitted²³. The ongoing development of such single-step homogeneous assays has further driven the miniaturization and automation of BLI reporter assays.

Both the BLI assay and the ⁵¹Cr assay show an E:T ratio-dependent proportional increase in cytotoxicity, but the BLI assay yields a greater signal-to-background ratio¹⁶ and an up to twofold higher percentage of specific lysis than the ⁵¹Cr assay^{16,24}. Rossignol et al. found a comparable signal-to-background ratio between a nanoluciferase assay and the ⁵¹Cr assay, with similar cytotoxicity curves but slight differences in half-maximum effective concentration (EC₅₀) values, likely because of the different release mechanisms of nanoluciferase and ⁵¹Cr (ref. ²¹).

Because of its ease of handling and simple quantification, the BLI assay has gained universal use (Table 3). However, a prerequisite for BLI-based reporter assays is the expression of luciferase in target cells. Stable transfection of cells is time-consuming and not feasible for every type of cell; therefore, the BLI assay is particularly useful for cases in which the same target cell line is used throughout the screening procedure. Of note, the BLI assay has also been successfully used with target cells that only transiently express the luciferase gene^{23,24}. Low transfection efficacies of plasmid-based approaches in primary cells can be overcome by electroporating target cells with luciferase gene-encoding RNAs produced in large amounts by *in vitro* transcription²⁵.

Impedance-based assay

A label-free alternative to the ⁵¹Cr assay and the BLI assay relies on microelectrodes embedded in the bottom of a microtiter well to measure the impedance of the flow of an electric current between electrodes upon adhesion of cells. Whereas impedance in the absence of cells is basically zero and is mainly dependent on the ionic environment of the medium, attachment of cells to the electrodes leads to a change in impedance that is dependent on the number, morphologic aspect (confluence), viability, and degree of adhesion strength for each cell type^{26,27}. Electrode impedance is reported as a ‘cell index’, a dimensionless parameter that is proportional to the number of adherent cells and is used to monitor changes in impedance over time. Suspension and nonadherent cells typically cause no change in electric impedance or significantly less change than adherent cells. This principle informs the measurement of cytolysis of adherent target cells by nonadherent effector cells, such as T cells, NK cells, and other immune-cell subsets²⁸. The workflow for this assay includes measuring the baseline impedance of medium alone, followed by seeding with the target cells. The seeding concentration of the target cells is optimized to reach a cell index in the upper to maximum range at the time that the effector cells are added, often 20–30 h after seeding, for reasons of convenience²⁹. In contrast to the ⁵¹Cr assay and the luciferase assay, the impedance-based assay allows label-free, real-time monitoring of cytolysis (measured as target cell

Table 2 | Applications of ⁵¹Cr-release assays for measuring cell-mediated cytotoxicity

Application	Effector cells/drug	Target cells	Target cell number per well	E:T ratio(s)	Incubation time (h)	Ref.
T-cell-mediated cytotoxicity	CD8 T-cell clone 3E6	U251T	2,000 or 5,000	50:1 to 0.3:1	4	24
	Human PBMCs and murine splenocytes	K562, U266, UCI01, P815, YAC1, EL-4, A20	100,000 ml ⁻¹	100:1 to 5:1	2, 4	16
	Mesothelin-targeted CAR T cells with intrinsic checkpoint blockade	MSTO-211H expressing human mesothelin	5,000	64:1 to 0.5:1	18	61
	TCR T cells specific for TARP	T2, Mel526 expressing TARP	1,000	50:1 to 0.4:1	4	81
	T cells recognizing influenza matrix protein-derived peptide	OAW42 loaded with peptide	20,000	5:1 to 0.005:1	6, 24	15
NK-cell-mediated cytotoxicity	HER2-specific T cells	SKBR3	100,000	40:1 to 1.25:1	5	34
	NK cells	K-562, Daudi	10,000	20:1 to 2.5:1	4	82
	NK cells	K562	Unspecified	100:1 to 3.125:1	4	44
Oncolytic virus-mediated cytotoxicity	NK cells, parvovirus H-1PV	H-1PV-infected Panc-1, MiaPaCa-2, Capan-1, CF-Pac1, AsPC-1	5,000	20:1 to 2.5:1	4, 16	83
BiTE-mediated cytotoxicity	γδ T cells, BiTE (HER2×CD3)	Panc89, Colo357	Unspecified	50:1 to 6.25:1	4	84
	Rituximab, T cells transduced with CD16	Autologous B-lymphoblastoid cell line	Unspecified	30:1	4	85
	Cetuximab, PBMCs	A549, N417, LK87, Lu99	10,000	40:1 to 10:1	4	86
	Rituximab, macrophages	Raji	Unspecified	50:1 to 12.5:1	8	87
	Rituximab or trastuzumab, T cells transduced with CD16	Raji, SKOV-3	3,000	10:1	4	21

Listed are examples of published assays to analyze cytotoxic effects mediated by different effector cells, illustrating the various uses of the ⁵¹Cr-release assay. ADCC, antibody-dependent cellular cytotoxicity; BiTE, bispecific T-cell engager; CAR, chimeric antigen receptor; E:T, effector to target; HER2, human epidermal growth factor receptor 2; NK, natural killer; PBMC, peripheral blood mononuclear cell; TARP, T-cell receptor γ chain alternate reading-frame protein; TCR, T-cell receptor.

Table 3 | Applications of bioluminescence assays for measuring cell-mediated cytotoxicity

Application	Effector cells/drug	Target cells	Target cells number per well	E:T ratio(s)	Incubation time (h)	Ref.
T-cell-mediated cytotoxicity	CD19-specific CAR T cells	Raji	Unspecified	10:1	4	23
	Murine splenocytes	Neuro2a	Unspecified	40:1 to 5:1	2 d	88
	CD8 T-cell clone 3E6, 2D7, 2A7	U251T, Daudi	2,000 to 10,000	50:1 to 0.3:1	1–4	24
NK-cell-mediated cytotoxicity	Human PBMCs and murine splenocytes	K562, U266, UCI01, P815, YAC1, EL-4, A20	300,000 ml ⁻¹	100:1 to 5:1	2, 4	16
	Splenocytes transduced with HER2-specific TCR	4T1 expressing HER2	Unspecified	20:1 to 5:1	Up to 72	20
	NK92MI with CD19-specific CAR	Raji	Unspecified	0.5:1	4	23
	NK cells	YAC-1, TS/A	5,000	40:1 to 10:1	4	89
	NK92 with EGFR-specific CAR, herpes simplex virus 1	MDA-MB-231	5,000	Unspecified	Up to 4 d	90
Oncolytic virus-mediated cytotoxicity	T cells, BiTE (EGFR×CD3)	SW480, OVCAR-8, A2058	2,500	10:1	48	91
	T cells, blinatumomab	Raji	Unspecified	20:1	4	23
	IMAB 362, PBMCs	KATO-III, NUGC-4	20,000 to 25,000	40:1	24	25
ADCC	Rituximab, PBMCs	Raji	Unspecified	40:1	4	23
	Rituximab or trastuzumab, T cells transduced with CD16	Raji, SKOV-3	3,000	10:1	4	21

Listed are examples of published assays to analyze cytotoxic effects mediated by different effector cells, illustrating the various uses of the bioluminescence assay. ADCC, antibody-dependent cellular cytotoxicity; BiTE, bispecific T-cell engager; CAR, chimeric antigen receptor; EGFR, epidermal growth factor receptor; E:T, effector to target; NK, natural killer; PBMC, peripheral blood mononuclear cell; TCR, T-cell receptor.

detachment) over an interval of time. Dynamic monitoring of treatment efficacy can be used to compare the cytotoxicity kinetics of different cells and varied constructs at multiple E:T ratios over a period of several days—a major advantage compared with other assays. Very low E:T ratios (in the range of 0.05:1 to 10:1) are feasible because of the nature of the assay, allowing it to be conducted over the course of several days^{15,30}, significantly reducing the number of effector cells needed.

As the measurement of impedance is dependent on the adherence of cells at the electrode–solution interface, precoating the wells with extracellular matrix proteins, such as fibronectin, laminin, and collagen, can improve the binding of lightly adherent cells and has been shown to induce adhesion of leukemia and lymphoma cells³¹. Another strategy to tether suspension cells to the electrodes is to coat the wells with antibodies against cell-specific surface markers (for example, CD19 or CD40 for B-cell tumors)³². Specialized assay applications exist to monitor cell migration or indirect cytotoxicity through soluble factors by using cell-permeable or cell-impermeable membranes, respectively³³.

Several groups have compared the impedance-based assay with the ⁵¹Cr assay, with inconsistent results. Peper et al. found a close correlation between the assays at 6 h and complete target cell lysis at 24 h for both assays, but the impedance-based assay exhibited a significantly higher percentage of specific lysis at E:T ratios <5:1 at 24 h (ref. ¹⁵). In contrast, Erskine et al. reported a significant difference between both assays at 5 h, with the impedance-based assay showing a higher level of lysis than the ⁵¹Cr assay³⁴. Whereas Erskine et al. used more than six times more target cells in the ⁵¹Cr assay, Peper et al. used the same number of target cells in both assays. Both groups concluded that the increased lysis in the impedance-based assay can be attributed to its higher sensitivity compared with the ⁵¹Cr assay. Owing to its ability to measure effector cell function at low E:T ratios without the need for manipulation of the target cells, the impedance-based assay is increasingly used to study immunotherapeutic interventions (Table 4).

Flow assay

Unlike the aforementioned assays, the flow assay allows for analysis and quantification of the cytotoxicity of cell subpopulations in heterogeneous cell mixtures, providing a unique advantage for the investigation of the differential treatment susceptibility of distinct cell types within a heterogeneous target cell population³⁵. The flow assay distinguishes between target and effector cells by their properties in terms of size and granularity (by forward and sideward scatter, respectively) and by specific staining of cells with antibodies coupled to a fluorescence dye. For the evaluation of cell death, DNA-intercalating fluorescent agents, such as propidium iodide or 7-aminoactinomycin D, which are preferably taken up by dead cells and undergo a spectral shift upon association with DNA, are commonly used³⁶. Stages of apoptosis can be assessed using cell staining with annexin V, which specifically binds to phosphatidylserine that is externalized during apoptotic cell death^{37,38}. Cell-permeable dyes that react with intracellular free amines, such as carboxyfluorescein succinimidyl ester, which is

frequently used in cell proliferation assays, can be used to further distinguish cell populations^{35,39}.

Reports of either target or effector cell staining in conjunction with a cell death or apoptosis marker have revealed a positive correlation between the ⁵¹Cr assay and the flow assay^{35,37,40,41}. Liu et al. developed a flow-based assay that measures cytotoxic T lymphocyte-induced caspase activation in target cells using a fluorogenic caspase substrate and reported that the flow assay was more sensitive than the ⁵¹Cr assay⁴². A similar approach was developed by Packard and Komoriya to measure cell death on the basis of activation of intracellular proteases using cell-permeable fluorogenic substrate probes. This method was used to quantify the entry of granzyme B into target cells as an early event in cell-mediated cytotoxicity⁴³.

Flow assays require the acquisition and analysis of individual sample data, which renders flow cytometry-based assays laborious unless an automatic workflow is implemented. Although the intra-assay variability was similar for the flow assay and the ⁵¹Cr assay, Motzer and colleagues found no compelling reason to adopt an NK-cell flow assay over the ⁵¹Cr assay, owing to the greater time requirement for the flow assay⁴⁴. However, a recently reported miniaturized and automated flow assay was found to be capable of reliably measuring NK-cell-mediated cytotoxicity in a 1,536-well format with a throughput of 50,000 wells per day⁴⁵. Owing to the ongoing development of high-throughput multiparameter instruments and mass cytometry at single-cell resolution^{46,47}, cytotoxicity and cellular behavior can now be studied concurrently in a timely fashion. Martinez et al. reported a high-throughput multiparameter flow assay that combines CAR T-cell-mediated killing with concurrent evaluation of CAR T-cell transduction and activation status⁴⁸. The capability of the flow assay to simultaneously analyze target cell killing and effector cell phenotype, as well as its ability to distinguish between heterogeneous cell populations, further advances the possibility of studying the dynamic relationship between costimulatory and coinhibitory signaling in a multicell environment in a single run. However, one must be cautious in experimental design, as well as the interpretation of multiparametric results beyond cytotoxicity by checking for reproducibility and confirmation by other assays.

Alternative assays

Other nonradioactive methods that are alternatives to the ⁵¹Cr assay include assays that directly measure target cell lysis, such as calcein- and europium-release assays. In the former, target cells are loaded with the lipid-soluble acetoxymethyl ester of calcein, which is converted by intracellular esterases to the lipid-insoluble fluorochrome calcein⁴⁹. Measurement of calcein in the supernatant is affected by autofluorescence of media components and a significantly higher spontaneous release of calcein, compared with ⁵¹Cr, leading to a low signal-to-background ratio⁵⁰. Consequently, advanced methods using high-throughput flow cytometry or cell imaging focus on the quantification of fluorescent live cells^{51,52}. In europium-release assays, the lanthanide europium, loaded into target cells in a complex with diethylenetriaminepentaacetate, is quantified in

Table 4 | Applications of impedance-based assays for measuring cell-mediated cytotoxicity

Application	Effector cells/drug	Target cells	Target cell number per well	E:T ratio(s)	Incubation time (h) after effector addition	Ref.
T-cell-mediated cytotoxicity	HER2-specific CAR T cells T cells recognizing influenza matrix protein-derived peptide	MC57 expressing HER2 OAW42	20,000 20,000	10:1 to 1:1 10:1 to 0.005:1	Up to 50 Up to 81	30 15
NK-cell-mediated cytotoxicity	HER2-specific T cells NK92	SKBR3 PC3, MCF7	7,500 10,000 to 40,000	40:1 to 125:1 10:1 to 0.625:1	Up to 20 >40	34 32
Macrophage-mediated cytotoxicity	Differentiated U937, ANA-1 Differentiated U937	Raji tethered to plate via anti-CD40 antibody MDA-MB-231, MDA-MB-435, MCF-7 MDA-MB-231	60,000 2,500 to 20,000 Unspecified	2:1 to 0.1:1 40:1 1:5	>20 48 72	32 92 93
Oncolytic virus-mediated cytotoxicity	Adenovirus-parvovirus chimera	HeLa, A549, ME-180, Lox-IMVI, HCT-15, HCC-2998, pMell	4,000 to 8,000	MOI = 100 to 1	Up to 1 week	94
BiTE-mediated cytotoxicity	Preactivated T cells, adenovirus expressing BiTE (EGFR×CD3) γδ T cells, BiTE (HER2×CD3; HER2× Vγ9) PBMCs, BiTE (Epcam×CD3)	A549, HCT116 PancTu-1, Panc89, Colo357 PC3	10,000 5,000 10,000	MOI = 1, E:T = 5:1 50:1-5:1 10:1 to 0.625:1	>90 Up to 120 >80	95 84 32
ADCC	NK-92, patient-derived mononuclear or polynuclear cells, trastuzumab	BT-474	5,000–20,000	1:1	72	96

Listed are examples of published assays to analyze cytotoxic effects mediated by different effector cells, illustrating the various uses of the impedance-based assay. ADCC, antibody-dependent cellular cytotoxicity; BiTE, bispecific T-cell engager; CAR, chimeric antigen receptor; E:T, effector to target; MOI, multiplicity of infection; PBMC, peripheral blood mononuclear cell; NK, natural killer.

Table 5 | Applications of flow cytometry assays for measuring cell-mediated cytotoxicity

Application	Effector cells/drug	Target cells	Staining	E:T ratio(s)	Incubation time (h)	Ref.
T-cell-mediated cytotoxicity	NY-ESO-specific T cells CD8 ⁺ T cells	Jurkat/A2K DohH2, Karpas, Raji	CFSE, CMTMR, propidium iodide CFSE, 7-AAD	1.5:1 to 0.1:1 50:1 to 2.5:1	4 24	97 98
NK-cell-mediated cytotoxicity	γδ T cells NK cells	THP-1 K562	CFSE, propidium iodide CellTrace Violet dye	20:1 to 1:1 1:1	4 5	39 45
Macrophage-mediated cytotoxicity	NK cells NK cells	K562 K562	Annexin V, propidium iodide, BCECF DiO, propidium iodide	1:1 to 1:10 75:1 to 6.25:1	1 to 3 3	99 44
Oncolytic virus-mediated cytotoxicity	NK cells U937	K562 IGROV1	DIOC18, 7-AAD CFSE, propidium iodide, CD89-phycoerythrin	20:1 to 1:1 2:1 to 1:1	4 2.5	100 101
BiTE-mediated cytotoxicity	PBMCs, adenovirus expressing BiTE (EGFR×CD3) CD3 ⁺ T cells enriched from PBMCs, BiTE (CEA×CD3)	A431, A549 CHO, ASPC-1, BxPC3, HPAC, HPAF II, H727, LS174T, HT-29, MKN45, Pan0813, A549, PC3, BT474	GFP, 7-AAD Vybrant DiO, propidium iodide	MOI = 20, E:T = 6:1 10:1 to 1:2	5 d Up to 72	95 102
ADCC	PBMCs, NK92MI, trastuzumab	BT-474, MCF-7, Hs578T	CFSE, FVD/7-AAD/propidium iodide	1:1 to 1:20	Overnight	103

Listed are examples of published assays to analyze cytotoxic effects mediated by different effector cells, illustrating the various uses of the flow cytometry assay. 7-AAD, 7-aminoactinomycin D; ADCC, antibody-dependent cellular cytotoxicity; BCECF, 2',7'-bis(carboxyethyl)-5(6)-carboxyfluorescein; BiTE, bispecific T-cell engager; CFSE, carboxyfluorescein succinimidyl ester; CMTMR, 5- and 6-((4-chloromethyl)benzoyl)-amino tetramethylrhodamine; E:T, effector to target; FVD, fixed viability dye; GFP, green fluorescent protein; MOI, multiplicity of infection; PBMC, peripheral blood mononuclear cell; NK, natural killer.

the supernatant of lysed cells upon transformation into a fluorescent chelate and detection of time-resolved fluorescence⁵³. Difficulties with target cell labeling accompanied by high spontaneous release of europium have prevented widespread use of this assay⁵³. Another class of cytotoxicity assays is based on cytosolic enzymes, such as lactate dehydrogenase, which are released from damaged cells and whose activity can be quantified via colorimetric or fluorometric readouts during the enzymatic reaction⁵⁴. One glaring limitation of this approach is that dead effector cells will also contribute to the enzyme concentration and measurement of cytotoxicity. As a result, it is suggested that, when using these assays, effector cell viability should remain above 95% to ensure the accuracy of measurements; however, events such as activation-induced death of effector cells may still confound the interpretation of results²³.

Another approach enumerates effector cell function by means of secreted proteins involved in specific pathways characteristic of cell-mediated cytotoxicity. The enzyme-linked immunospot assay operates under a principle similar to the classic enzyme-linked immunosorbent assay to measure the release of effector cytokines during effector and target coculture⁵⁵. Efforts to quantify effector cell responses have centered around interferon-gamma (INF- γ)⁵⁶ as a measure of effector cell activation, as well as granzyme B and perforin as key mediators of cell death via the granule-mediated pathway^{57,58}. Granzyme B and perforin may be more representative surrogate markers for cell-mediated cytotoxicity than INF- γ , as INF- γ is not always found to be secreted by cytotoxic lymphocytes⁵⁸. Degranulation of T and NK cells can be further assessed by CD107a flow cytometry⁵⁹. CD107a becomes accessible to antibody staining during effector cell degranulation, and its expression is correlated with target cell lysis⁶⁰. It is critical to emphasize that this class of assays assesses only effector cell function as an indirect measure of cell-mediated cytotoxicity, whereas assays such as the ⁵¹Cr assay directly measure target cell death.

Advanced in vitro antitumor efficacy assays

Whereas in vitro efficacy assays typically investigate treatment modalities under single-effector, single-target, or single-antigen conditions, additional layers of complexity can be introduced to mimic the conditions that CAR T cells and other cell therapies face in the tumor microenvironment in vivo. Inhibition of cytolytic and cytokine secretion functions resulting from chronic antigen exposure-induced T-cell exhaustion can be investigated during repeated antigen stimulation. In this ‘antigen stress test’, effector cells are harvested and repeatedly transferred to wells with freshly seeded target cells at a defined E:T ratio. Retention or loss of effector function upon repeated antigen stimulation can be quantified using the assays outlined above to compare, for example, the effects of different costimulatory domains on effector cell-mediated cytotoxicity under high antigen stress^{4,14,61}. With the modification of variables such as E:T ratio, target antigen exposure time, the measurement of intermittent cytotoxicity following specified rounds of antigen exposure, and the addition of rescue

immunomodulatory antibodies, antigen stress tests can yield more comprehensive information than a single-exposure cytotoxicity assay.

Unlike CD19, which is specifically used as a target antigen for tumors of the B-cell lineage, solid tumor antigen targets are expressed heterogeneously⁶². This variable expression and distribution of antigens constitutes a hurdle for cell therapy, as it potentially renders tumor cells with low antigen density less susceptible to treatment and could increase the risk of ‘on-target/off-tumor’ toxicities toward normal tissue with very low expression of the target antigen⁶³. Assessing the differential cytotoxicity of CAR T cells toward targets with variable antigen expression provides an opportunity to investigate the antigen activation threshold, the correlation between antigen density and cytotoxicity, the additive cytotoxicity on low-antigen targets in the presence of high-antigen targets, and on-target/off-tumor toxicity. In a similar fashion, simultaneously targeting more than one antigen, either by using dual-antigen CAR T cells or by coadministering two or more CAR T-cell constructs (each with single-antigen specificity), represents a strategy to overcome tumor-associated antigen heterogeneity and tumor immune escape⁶³. Differential cytotoxicity can be assessed through selective labeling of the target cell in a heterogeneous mixture of cells in the ⁵¹Cr assay, or by transduction of target cells with different fluorophores in the flow assay.

In addition to facing antigen heterogeneity, CAR T cells encounter varied immune cells and tumor cell-secreted inhibitory or excitatory cytokines within the tumor microenvironment. In particular, regulatory T cells and M2 macrophages, among others, can inhibit T-cell function⁶⁴. Moreover, soluble factors and cytokines, such as transforming growth factor beta (TGF- β), play a role in modulating treatment response⁷. The inclusion of soluble and cellular components into the assay system allows investigation of CAR T-cell efficacy in the presence of immunomodulatory factors. Our laboratory has developed a malignant pleural effusion-derived ex vivo culture system to investigate the efficacy of immunotherapeutic agents, including CAR T cells derived from a patient’s own T cells, in a human, immunocompetent, tumor-like environment⁶⁵ and to further test the ability of immune modulatory antibodies against PD-1 or TGF- β to rescue CAR T-cell cytotoxicity.

The advanced assays described above allow the investigation of the cytotoxicity of cell therapies under conditions resembling the environment these living drugs face in the tumor microenvironment in vivo. Table 6 provides an overview of the assays discussed in this section. More importantly, after gaining expertise in the conduct and interpretation of the assays described above, investigators can rationally combine the assays to simultaneously investigate transduced T-cell expansion, memory, and exhaustion phenotypic changes and effector cytokine secretion, in addition to cytotoxicity. For example, while conducting repeated antigen stress tests with multiple effector T cells against heterogeneous antigen expression targets, in addition to cytotoxicity at predetermined time points, one could also assess transduced and un-transduced T-cell expansion by cell count or bead assay, phenotypic changes by flow cytometry, and effector cytokine secretion by analysis of supernatants with Luminex assay. A combined interpretation of

Table 6 | Advanced in vitro antitumor efficacy assays

Assay	Investigation of focus	Description	Recommended assay(s)
Antigen stress test	Adaptive resistance, cytotoxicity following functional exhaustion	Repeated antigen stimulation by multiple rounds of target cell addition to investigate effector cell cytotoxicity under high antigen stress	⁵¹ Cr, BLI, impedance, or flow cytometry assay ^a
Differential cytotoxicity on target cells with heterogeneous antigen expression	Cytotoxicity on target cells with antigen heterogeneity (low and high), on-target/off-tumor toxicity on normal cells with very low antigen expression	Cytotoxicity on target cells with variable antigen expression to investigate the correlation between antigen density and cytotoxicity and any unanticipated cytotoxicity of activated effector cells on normal cells with very low antigen expression	If target cells with different antigen expression levels are assessed individually (one cell type per well): ⁵¹ Cr, BLI, impedance, or flow cytometry assay. If target cells are plated as mixture: flow cytometry assay
Multi-CAR cytotoxicity	Antigen heterogeneity, antigen escape	Cytotoxicity assessment of effector cells with multiple CAR constructs targeting different antigens simultaneously	⁵¹ Cr, BLI, impedance, or flow cytometry assay
Cytotoxicity in the presence of soluble factors	Immunosuppression by soluble factors known to be present in the tumor environment	Cytotoxicity in the presence of various doses of single or multiple inhibitory or excitatory cytokines and/or soluble factors	⁵¹ Cr, BLI, impedance ^b , or flow cytometry assay
Cytotoxicity in the presence of immune cells	Immunosuppression by immune cells known to be present in the tumor immune environment	Cytotoxicity in the presence of various ratios of single or multiple immunomodulatory immune cells	Flow cytometry assay to investigate cytotoxicity against heterogeneous (target) cell populations

^aIt is imperative that the target cells are completely lysed before the effector cells are pooled from the coculture condition and exposed to freshly plated target cells. Transfer of viable target cells to the next round of antigen stimulation might change the E:T ratio, potentially confounding comparison across different effector constructs. ^bAddition of soluble factors to the impedance assay may impact tumor cell morphology and cell index. It is critical to include a control to account for soluble factor-induced changes in cell index in the absence of effector cells. As an example, treatment of target cells with soluble factors such as TGF- β or tumor necrosis factor- α can alter the size, adherence, and cell surface receptor expression on target as well as T cells, and it is important to include controls at different concentrations and durations of incubation. CAR, chimeric antigen receptor

the results from these assays provides insights into underlying biological changes in addition to cytotoxicity⁶¹.

Potency assays for clinical cell therapy products

The potency requirements for investigational cell therapy products vary with the phase of investigation, and although a potency assay is not required to initiate early-phase clinical studies, a qualified assay to determine the dose is recommended⁶⁶. A validated in vitro or in vivo assay that measures the biological activity is required by the end of phase 2 (ref. ⁶⁷). Due to the complexity of the mode of action of cellular products, no standardized potency assays exist, but the assay should represent the product's relevant biological properties⁶⁸. Although characteristics such as migration, proliferation, persistence, and specific phenotypes are linked to positive clinical outcomes of adoptive cell therapies, only the cytotoxic activity of the effector cells leads to a reduction in tumor burden and is therefore considered the preferred mode of action for a release assay⁶⁹.

Despite the vast diversity of cytotoxicity readouts available and presented in this article, the majority of CD19-targeting CAR T-cell therapies in clinical trials utilize the ⁵¹Cr assay and INF- γ release as potency assays for product release⁷⁰. Fewer studies utilize flow cytometry to determine the percentage of live target cells after coculture with the final product^{71,72}. Potency assays for the two FDA-approved CD19-specific CAR T-cell products (tisagenlecleucel and axicabtagene ciloleucel)

measure the percentage of transduced cells and INF- γ production in response to CD19-expressing tumor cells^{73–75}. However, an FDA briefing document for tisagenlecleucel states that INF- γ production varied greatly from lot to lot in the clinical trials, complicating the correlation of INF- γ production to product safety or efficacy⁷⁴. With emerging changes in cell manufacturing, such as the addition of interleukin (IL)-15 or IL-21 or selecting a phenotype, these products are to be validated in vivo (in addition to the above-mentioned in vitro assays) before entering the clinic.

Assessment of cell therapies beyond cytotoxicity

Successful immunotherapy is not only based on the cytotoxic activity of engineered immune cells toward the tumor target but also includes tumor infiltration and migration, antigen-dependent expansion, cytokine secretion, functional persistence, alleviation of exhaustion, and the recruitment of the innate and adaptive immune system. Therefore, to assess cell therapies in vitro and in vivo, investigators rely on additional assays⁷⁶.

Antigen-dependent T-cell expansion can be measured by fold-increase in cell number or number of cell divisions. The latter is typically achieved by labeling the effector cells with a fluorescent dye that is equally split between the daughter cells, resulting in a diminished fluorescence with each cell division⁷⁷. Persistent exposure to antigen can lead to exhaustion, a state characterized by loss of effector cell function, that is

accompanied by an increase in inhibitory receptors such as PD-1, whose expression can be determined by flow cytometry. Furthermore, the differentiation status and certain memory features of the effector cells are associated with antitumor efficacy and persistence. The percentage of memory T cells and terminally differentiated T cells can be measured by staining for CD45RA and CD62L, among other markers⁷⁸. Ultimately, functional persistence, an indicator of long-term tumor control, can be assessed by performing tumor rechallenge experiments in vivo¹⁴. Although certain characteristics mentioned above can be established concurrently during repeated antigen stimulations, we encourage the investigator to use orthogonal assays to gather additional performance parameters, considering the specific cell therapy platform utilized and its unique mode of actions.

Discussion

The emerging role of genetically engineered immune cells, oncolytic viruses, and immunomodulatory agents for the treatment of cancer and autoimmune diseases relies upon sensitive and robust in vitro potency assays for the research, development, and manufacturing phases of these living drugs. These assays involve (1) labeling of the target cells (such as ⁵¹Cr, live or dead, or antibody staining for markers of interest in flow cytometry), (2) reporter enzyme activity (e.g., luciferase and green fluorescent protein (GFP)), (3) cell attachment or detachment (impedance-based assay), or (4) release of compounds from effector cells (e.g., IFN- γ and granzyme B). The underlying physical and biological mechanisms for quantifying drug potency and measuring cell-mediated cytotoxicity differ substantially across the platforms. Whereas, for example, INF- γ - and granzyme B-based assays focus on effector cell function, the ⁵¹Cr assay measures target cell death. Since effector cell function does not always relate to target cell death, this review focuses on assays that directly measure target cell lysis.

Owing to the inherent variability in function between different donor effector cells, there are several requirements for cytotoxicity assays to be versatile tools for the study of immunotherapeutic agents. They should ideally possess (1) high specificity and sensitivity, (2) the ability to reflect the mode of action, (3) linearity over a wide concentration or E:T range, (4) robustness, (5) reproducibility, (6) scalability, and (7) automatability. Although the ⁵¹Cr assay has historically been the gold-standard measurement of cell-mediated cytotoxicity, powerful technologies for cytotoxicity assessment that do not rely on radioactive labeling of target cells and provide comparable results, with increased sensitivity and higher sample throughput, have now been developed. As such, the traditional ⁵¹Cr assay is starting to lose its primacy. Its signal-to-background ratio and sensitivity have been reported to be inferior to those of the BLI assay¹⁶, impedance-based assay¹⁵, and flow assay⁴². A certain limitation of the ⁵¹Cr assay is its restriction to measure time points within ~24 h, after which specific ⁵¹Cr release usually becomes indistinguishable from background release¹⁵. In addition, the high E:T ratios required in the ⁵¹Cr assay are not representative of the ratios observed in patients with cancer, who often have a high tumor burden and

relatively low numbers of effector cells infiltrating into the tumor^{79,80}. The BLI assay, impedance-based assay, and flow assay allow for the application of much lower E:T ratios that are physiologically more relevant.

With our laboratory's focus on solid tumor CAR T-cell therapy translation, we perform an impedance assay that allows repeated antigen stimulation and assessment over at least 4 d following initial ⁵¹Cr assay and a focused flow assay based on the results of the first two assays. Each assay platform has intrinsic strengths but also intrinsic weaknesses (Fig. 1). The decision of which assay to use should be based on the individual properties of the experiment. One should consider the characteristics of the target and effector cells (e.g., primary cells, cell line or cell mixture, adhering or suspension cells, easy- or hard-to-transfect cells), as well as the number of samples to be measured and the level of automation required. The impedance-based assay is unique in that it does not require any cell labeling. The flow assay has the distinct advantage of allowing the analysis of differential cytotoxicity in a heterogeneous cell population. Such characteristics set these assays apart from the ⁵¹Cr assay and the BLI assay when it comes to hard-to-transfect cells and multiparameter readouts, respectively. Whereas the ⁵¹Cr assay is laborious and not suitable for high-throughput applications, the impedance-based and the BLI assays can be scaled up to high-density screening formats. Although it has historically been hindered by laborious sample preparation and slow data acquisition, the flow assay can be implemented in a high-throughput workflow dedicated to accelerating phenotype-based drug discovery⁴⁵.

Cytotoxicity assays compatible with downstream applications and automated workflows, such as the BLI, impedance-based, and flow assays, have the capacity to accelerate the discovery of novel living drug modalities in a high-throughput manner. With increased knowledge of the agents being used to augment synthetic immunity, these assays can aid in the comprehensive assessment of immune-cell function.

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Additional information

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